

Designing systems to monitor carbon stocks in forests and shrublands

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Abstract

The United Nations framework convention on climate change (FCCC) commits signatory nations to monitor changes in all fluxes and sinks of carbon, including those related to vegetation and soil. This paper describes a system for monitoring carbon in New Zealand's forests and shrublands (6.3 and 2.6 million ha, respectively), which was tested on a 60 km-wide transect across the South Island. A 9 km² grid was superimposed onto a land-cover map (SPOT imagery) to obtain 62 ground-sampling points. New permanent plots were established at 43 of these points and 18 existing plots were revisited (one site was inaccessible). On each plot, the dimensions of all trees, shrubs and coarse woody debris (CWD) were measured, and these measurements converted to per-hectare C-stocks using regression relationships. C-stocks in fine litter and mineral soil were quantified using data from New Zealand's national soil-C monitoring system that quantifies national-scale soil-C stocks using a soil-C database stratified by soil type, climate, and land-cover; we used these data to derive values for the transect.

The total C-stock in all pools was 290 t/ha in forests and 163 t/ha in shrublands, 58 and 30% of which was living biomass (respectively). The living biomass components had the greatest variances, so precision could be improved by increasing the sample size of trees, particularly large trees. The C-stock of CWD was highly variable among plots, but was a relatively small component of the total C-stock (10% forest and 3% shrubland). For the moment, we know little about the uncertainty associated with measuring C-sequestration (i.e. change in C-stocks over time), but this information will become available when plots are resampled.

Our network of permanent plots is designed deliberately to be simple and measurements of interest to conservation biologists, such as species composition and structure are included in the hope of protecting it against inevitable shifts in policy and focus. We discuss methodological issues including generalized allometric relationship, the propagation of errors, and better integration of remote-sensing and ground-based approaches, all of which have been considered in the development of this system. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

The amount of carbon in terrestrial ecosystems far surpasses that in the atmosphere (cf. 2000 versus

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670 Gt), so mankind has the potential to greatly influence atmospheric CO₂ concentration through management of these ecosystems (Bouwman, 1990; Sarmiento et al., 1995; Fan et al., 1998). Much attention has focused on changes in forest cover around the world, particularly the alarming rates of deforestation in many tropical countries (Laurance, 1998; Kauffmann-Zeh, 1999). Remote-sensing is suitable for the monitoring of such land-cover changes over large spatial scales (e.g. Skole and Tucker, 1993; Foody et al., 1997; Saatchi and Rignot, 1997; Cihlar, 2000). However, the remote-sensing techniques needed to reliably estimate per-hectare C-stocks are still in their infancy (e.g. Steininger, 2000), so ground-based sampling is needed to monitor these changes.

Evidence is accumulating that many terrestrial ecosystems are not at biomass equilibrium, even when averaged over large spatial scales. For example: (a) forests in eastern USA continue to accumulate biomass, more than a century after the abandonment of farming by European settlers (Foster, 1992; Schroeder et al., 1997; Fan et al., 1998); (b) humans influence the frequency of burning in many dry tropical forests, Mediterranean shrublands, and boreal forests, thereby influencing mean per-hectare C-stocks (e.g. Bowman, 1998); (c) invasive trees such as *Myrica faya* in Hawaii, and *Acacia* spp. in South Africa are replacing low-growing native vegetation (Vitousek et al., 1997); (d) climate change and enhanced CO₂ concentrations may have direct physiological effects on tree growth rates (Malhi and Grace, 2000) and indirect effects on the rate of mortality and turnover (Phillips, 1996); and (e) introduced animals may change forest structure and composition. The last point is exemplified in New Zealand by introduced possums (*Trichosurus vulpecula*), which are implicated in the dieback of *Metrosideros umbellata* forests, and deer which reduce sapling numbers and may cause regeneration failure (Wardle, 1991). In all these examples, the historical and present day actions of mankind are the cause of changes in per-hectare C-stocks.

The United Nations framework convention on climate change (FCCC) requires parties to report on all sources and sinks of carbon, including vegetation and soil. In addition, signatories of the Kyoto protocol (from Annex 1 countries) will be obligated to monitor 'the net changes in greenhouse gas emissions ...

resulting from direct human-induced land-use change and forestry activities' although there is still uncertainty about which carbon stocks need to be accounted for, or the degree of precision required.

There are already many published estimates of carbon stocks in vegetation at a range of spatial scales, but we are not aware of any based on plot data specifically intended to provide nationally unbiased estimates of C-stocks. Most studies use data collected on plots that were selected to be typical of a location, rather than using a statistical sampling method (Brown and Gaston, 1995; Delaney et al., 1997, 1998). The tendency for such plots to have been established in relatively undamaged or late-successional forests means that they tend to overestimate live woody biomass (Brown et al., 1991; Brown and Lugo, 1992) and give disputable values for carbon flux (e.g. Phillips et al., 1998). Another source of data is national forest inventories, such as those conducted in the USA (Brown et al., 1997; Schroeder et al., 1997), Sweden (Hägglund, 1985), and Finland (Tomppo, 1996a). These data were collected using robust sampling approaches, sometimes over many decades (Kolchugina and Vinson, 1995), but were not originally intended to provide estimates of carbon stock and fluxes, and lack information on coarse woody debris (CWD), root allocation, soils, and small trees (Schroeder et al., 1997). Several countries are supplementing their inventory data with additional measurements, including expansion factors (as used in the IPCC default approach). Data from shrublands are uncommon around the world, but shrublands may need much closer attention if reversion of agricultural land to native vegetation is included as a C-sink under the Kyoto protocol. Whatever the final rules are, there is a need to design internationally acceptable systems that are capable of delivering unbiased and verifiable estimates that include a full range of components at relevant scales and with known precision.

The purpose of this paper is to contribute to the international scientific debate about how to design inventory systems that provide verifiable estimates of carbon stocks in forests and shrublands (Coomes and Beets, 1999). We describe a proposed monitoring system for New Zealand that uses a combination of ground-based plot measurements and land-cover area estimates from satellite imagery. The rationale for this monitoring system is discussed in terms of its

evolution from established databases, and its testing on a 60 km-wide transect across New Zealand's South Island. We emphasize the issues that have arisen during the course of its development, many of which we believe will be shared with other countries. We describe how the various carbon components can be measured and integrated to give a more complete estimate than has usually been reported. There is a school of thought that says that ground-based sampling can be used to construct regression models that reliably predict C-stocks as a function of climatic, edaphic, topographic, and remotely sensed variables. We therefore discuss whether the need for extensive ground-based monitoring will diminish once these relationships are defined. The network is designed primarily for carbon reporting requirements, but also as a basis for national and international reporting (e.g. state of the environment, montreal process, convention on biological diversity), and we briefly describe additional plot measurements aimed at providing meaningful data to other end-users. The system described is not specifically designed to meet reporting needs under the Kyoto protocol (vegetation change resulting from direct human activity) although many issues of design are relevant to that.

2. Methods

2.1. *Evolution of a monitoring system from a 1990-baseline study*

The intent of the Kyoto protocol is to set legally binding targets emission reduction, based on 1990 emission levels. Each nation is establishing 1990-baseline values from existing data (Cannell and Milne, 1995; Cannell et al., 1996). The 1990-baseline estimate of C-stock in New Zealand's indigenous forests and shrublands (Hall et al., 2001) was based on per-hectare C-stocks derived from permanent plot data stored in the National Vegetation Survey (NVS) databank (Wiser et al., 2001), and land-cover from the vegetation cover map (VCM) of Newsome (1987), and identified several components of the C-inventory that needed improvement (Allen et al., 1998). It identified that the NVS databank only contained information on live-stems, and that few suitable data had previously been collected in shrublands, and CWD had been

measured in only a few locations (e.g. Stewart and Burrows, 1994). It also recognized that a monitoring system could not rely entirely on revisiting existing NVS plots. Although the 11,000 NVS plots in forests were randomized at a catchment level, the sampling is most intensive in mountainous regions, and under-represents some forest types (Fig. 1). Therefore, a truly nationwide network of permanent plots was proposed, which would encompass shrublands as well as forests, and include systematic measurements of CWD. National C-stocks in living biomass and CWD would be monitored by: (a) measuring the dimensions of trees, shrubs and CWD in a network of permanent plots; (b) converting these dimensions into per-hectare C-stocks using regression relationships; and (c) multiplying these C-stocks by the spatial area of these vegetation types, obtained from satellite imagery.

An analysis of variance using existing NVS data on live-stem biomass established that around 500 plots (each of 0.04 ha) would be needed nationally to estimate carbon stock in forest with a 95% confidence interval that lay within 5% of the mean, and that a 9 km² grid superimposed onto the VCM would produce the requisite number of plots (Allen et al., 1998).

This led to a pilot study to test the logistics, and to identify the shortcomings, of such a national plot-based carbon monitoring system. The pilot study focused on an east–west transect of New Zealand's South Island, and has the 42°45NS and 43°15NS parallels as its northern and southern boundaries (Fig. 1). The 260 km-long transect passes through intact rainforest on the West Coast (rainfall >8000 mm per year), through the southern Alps (>2000 m a.s.l.), to human-induced shrublands in the east (rainfall cf. 600 mm per year). Hence, the pilot study was conducted across a broad range of forest types and disturbance conditions (Wardle, 1991). The transect area includes country with difficult access, therefore field procedures were chosen so that a sample point represented no more than one full day's field work, for a party of two to four persons.

2.2. *Land-cover database and selection of sampling points*

Since the 1990-baseline work was completed, a satellite-based land-cover database (LCDB) has largely replaced the VCM, which was based on aerial

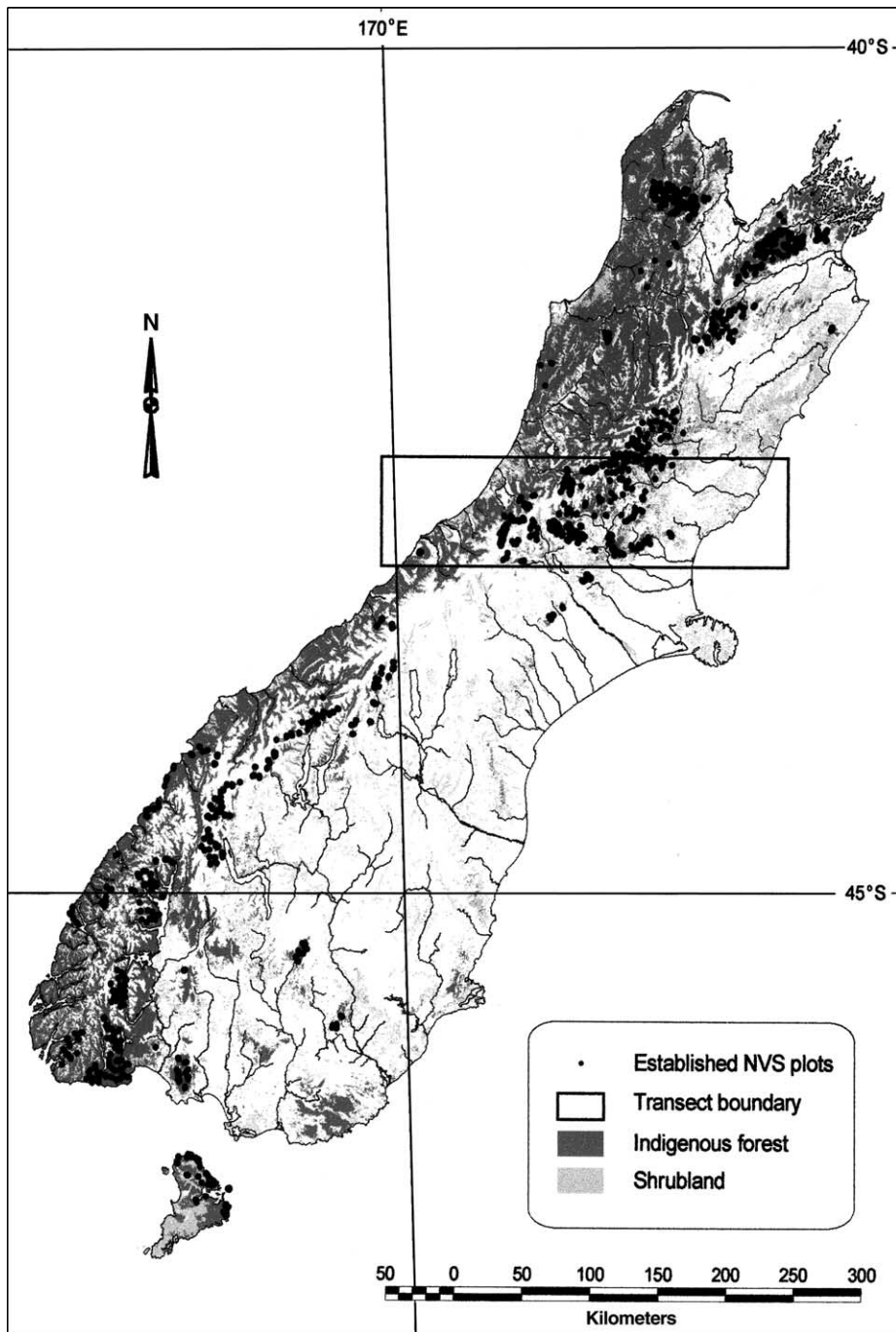


Fig. 1. South Island of New Zealand, showing locations of permanent NVS forest plots, and the position of the transect used in this study.

photography and extensive ground-truthing. The LCDB was derived from a classification of SPOT satellite imagery acquired in the summer of 1996/1997. With an image pixel size of 20 m H 20 m, the spatial resolution was set at 1 ha (50 times higher than the VCM). The LCDB categorizes native vegetation as 'woody' when woody vegetation covers >20% of a 1 ha pixel. Woody vegetation is in turn categorized as a shrub when predominantly <6 m in height, and forest when predominantly >6 m in height. By superimposing a 9 km² grid onto a LCDB map of the transect, 39 sampling points were produced in indigenous forest, and 23 in shrubland (Fig. 2). In recognition of the value of long-term permanent plots, it was decided that an existing NVS plot would be used whenever one or more was located within the 9 km square centered on a sampling point (18 of 62), and that new plots would be established otherwise. Where there was more than one existing NVS plot, the nearest plot was selected. Exhaustive checking of the NVS database demonstrated that this sampling approach did not introduce detectable bias (Hall et al., 1998, unpublished data).

When (if) a national monitoring system is established, it may well include sample taken from vegetation types other than forest and shrub such as non-intensive farmland. This extension is important, as it will allow spread/shrinkage of shrublands and forests to be accurately monitored, and provide tests of the accuracy of the LCDB classifications.

2.3. Ground-based sampling for living trees in forest plots

Plot coordinates were marked on topographical maps, and then located using a combination of traditional methods (i.e. compass and tapes), and GPS where possible. When the field party reached a new sampling point, they classified the surrounding hectare of land as forest, shrubland or non-woody vegetation, using the LCDB definitions. If the area was forested, a permanent plot was established following a standard protocol based largely on standard inventory techniques. The forest plots had a nested design, in which large trees were measured in large plots and in trees in smaller plots. Trees greater than 60 cm were tagged and measured within a circular plot of 20 m radius. A plot of 20 m H 20 m was centered on the

circle, within which all trees greater than 2.5 cm diameter were tagged at breast height (1.4 m). These trees were identified by species, and their diameters were measured at breast height (dbh). In addition, the heights of 20 trees (selected non-randomly to represent the range of sizes present) were measured with a vertex hypsometer (Haglöf, Sweden). The heights of the remaining trees were estimated from height–diameter relationships obtained by least-squares regression.

For completeness we mention the other ecological measurements (not directly related to carbon) were taken within the 20 m H 20 m plot, using standard NVS methods (Allen, 1993). The plots were subdivided into 16 subplots of 5 m × 5 m, within which saplings were counted by species, while seedlings with 24 circular plots (49 cm radius) were counted by species in three tiers. A list of all vascular plant species (with cover classes) was obtained on each plot.

The total above-ground biomass (AGB) of each tree was estimated using the following relationship:

$$\text{AGB} = 0.0000598\rho(d^2h)^{0.946}(1 - 0.0019d) + 0.03d^{2.33} + 0.0406d^{1.53} \quad (1)$$

where h is tree height (m), d diameter at breast height (cm), and ρ species-specific basic wood density (kg/m³), taken from wood specimens archived at Landcare Research. The relationship was derived from data collected by Beets (1980), who destructively sampled 14 canopy trees and 36 subcanopy trees in a temperate rain forest (western South Island) and used generalized linear modeling (GLM) (assuming a gamma error distribution and log-link function) to obtain power relationships for stem wood, branches and leaves (Hall et al., 1998, unpublished data). The first term in Eq. (1) is stem wood mass, with an accommodation for hollow trunks ($1 - 0.019d$), while the second or third terms are branch and leaf components giving a parameters relationship.

The AGB of each tree less than 60 cm dbh was summed in the square plot, and the AGB of each tree greater than 60 cm dbh was summed in the larger circular plot. We added 25% to accommodate the dry mass of roots (i.e. 20% of total mass, Cairns et al., 1997). We are in the process of obtaining new relationships by sampling trees from a larger number of sites and species, and including more extensively

collected wood density data and root biomass data, these are issues we revisit in the Section 4. The biomass of large and small trees were expressed in t/ha, then dry mass was converted to a C-stock using a conversion factor of 0.50. Finally, the C-stock was corrected for the average slope of the plot by dividing the estimate by $\cos(\text{slope})$.

2.4. Ground-based sampling for live shrubs in shrubland plots

Shrub communities are challenging because they exhibit a variety of forms, ranging from individual single- to multiple-stemmed bushes, to intermingled plants, sometimes with very large numbers of individuals per hectare. Also, the distinction between a shrub and a tree is largely artificial, based on a 6 m height threshold. Standard methods for measuring shrub biomass were not as well-established in New Zealand, so a protocol was developed on techniques described in Catchpole and Wheeler (1992).

The same plot establishment procedure, size and shape were used for shrublands as for forest (i.e. a 20 m H 20 m), thereby allowing straightforward comparisons of different vegetation types. Any tree (>6 m height) found in a plot having a diameter (at breast height) of more than 2.5 cm was tagged, and treated in the same manner as trees in the forest plots. The field-crew had to decide at each plot whether the more appropriate method was to measure individual shrub volumes or take cover/height estimates on the following basis. Measuring individual volumes was the preferred option whenever shrubs were sufficiently distinct; the cuboid volume of each shrub was estimated from the two orthogonal widths and height of an imaginary rectangular box within which the shrub just fitted. Cover/height measurements were taken at 25 sampling points (at the corners of each 5 m × 5 m) (subplot) in plots where shrubs were highly intermingled. Cover was estimated by eye as a proportional cover of woody vegetation in a 20 cm × 20 cm centered on the sampling point. The cuboid volume of shrubs in the surrounding 5 m × 5 m was then estimated as 5 × 5 × height × cover.

These non-destructive plot measurements were converted to C-stocks by regression relationships established by the following methods. For plots where individual shrubs had been identified and measured,

three shrubs were selected objectively at a predefined distance away from the plot, their widths and height taken, and cuboid volumes calculated. The shrubs were then clipped to ground level. For plots where cover/height measurements had been taken, a strip 4 m × 1 m was selected at a predefined distance from the plot edge, its height and proportional cover measured, and cuboid volume calculated. The vegetation within the strip was clipped to ground level. In both instances the fresh weight of clipped material was measured in the field using a spring balance, and a sub-sample was taken after the harvested vegetation had been chopped and thoroughly mixed. The sub-sample was taken in a sealed polythene bag to a laboratory where it was oven dried at 65 °C to constant weight. An average moisture content was obtained and the density (oven dry weight per unit cuboid volume) of each shrub sample was then calculated as:

$$\text{shrub density} = (1 - \text{moisture content}) \times \frac{\text{fresh weight}}{\text{cuboid volume of sample}} \quad (2)$$

By collecting data from each shrub plot in the transect, a dataset of species-specific densities was obtained. An analysis of variance on logged data was used to obtain the mean densities of species sampled. Other species (recorded on the 0.04 ha plots but not destructively sampled) were given the average density value. The total AGB of shrubs in a plot was then obtained by summing the masses of all shrubs present:

$$\text{AGB} = \sum (\text{shrub density} \times \text{cuboid volume}) \quad (3)$$

In plots where cover/height measurements were taken, destructive harvesting provided a representative composite shrub density for the neighboring plot. The 25 estimates of cuboid volumes were summed (each representing a 25 m²) and multiplied by the plot-average shrub density, to give the total AGB in a square of 625 m².

The above-ground mass of any trees in the plots were added to the shrub masses. As for forested plots, the total dry mass of shrubs in a plot was obtained by adding 25% to the AGB to accommodate the dry mass of roots. The dry mass was then converted to a C-stock using a conversion factor of 0.50, expressed in t/ha, and slope corrected.

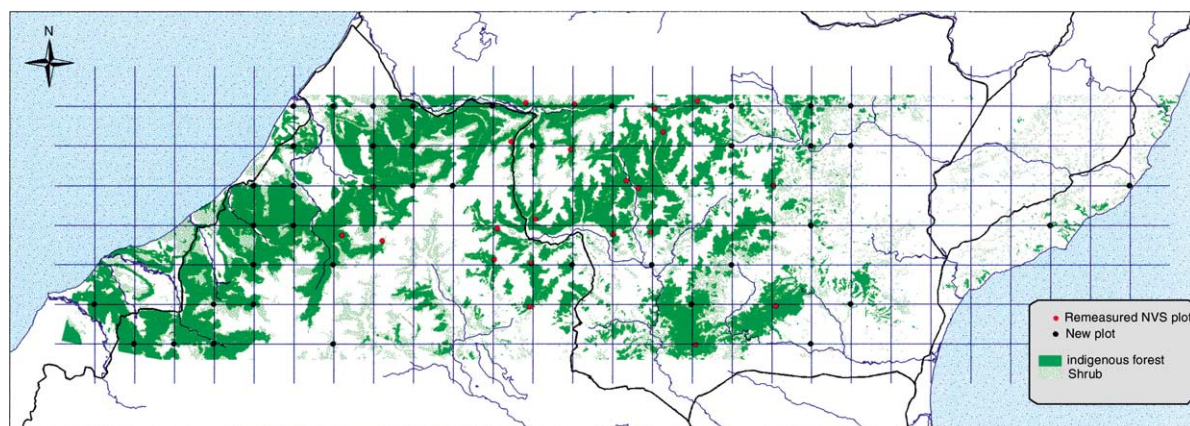


Fig. 2. A satellite-based vegetation map (LCDB) of a transect across South Island, New Zealand, onto which a 9 km² grid is superimposed. New permanent plots were established at the grid intersections wherever forest or shrubland was identified on the LCDB (black dots), unless a existing permanent plots was located within the vicinity, which was selected for remeasurement (red dots).

2.5. C-stock in coarse woody debris

The volumes of all fallen logs and standing dead trees greater than 10 cm diameter were measured within the 20 m H 20 m plots. Line intersect sampling is widely used in New Zealand for assessing deadwood and debris following logging, and is significantly more efficient than a sample of bounded plots over a wide area (De Vries, 1973; Van Wagner, 1968; Warren and Olsen, 1964). However, full enumeration was the preferred method within the 20 m H 20 m NVS plot, because some plots contained the majority of their CWD debris in a few large, misshapen and partially decayed logs (Stewart and Burrows, 1994). It

was deemed more appropriate to carefully measure all logs than to marginally reduce within-plot measurement time, incurring the risk of not counting them, and increasing sampling error. Full enumeration also enables tracking of standing trees through time should they fall or be blown over.

Standing dead trees and logs were identified by species whenever possible, and were categorized into one of three broad decomposition classes: (a) bark largely intact; (b) bark and twig lost, but shape of trunk remaining intact; and (c) shape no longer maintained, and trunk sinking into ground. Two measurements of diameter were made at right angles to one another at both ends of each log, and length was measured.

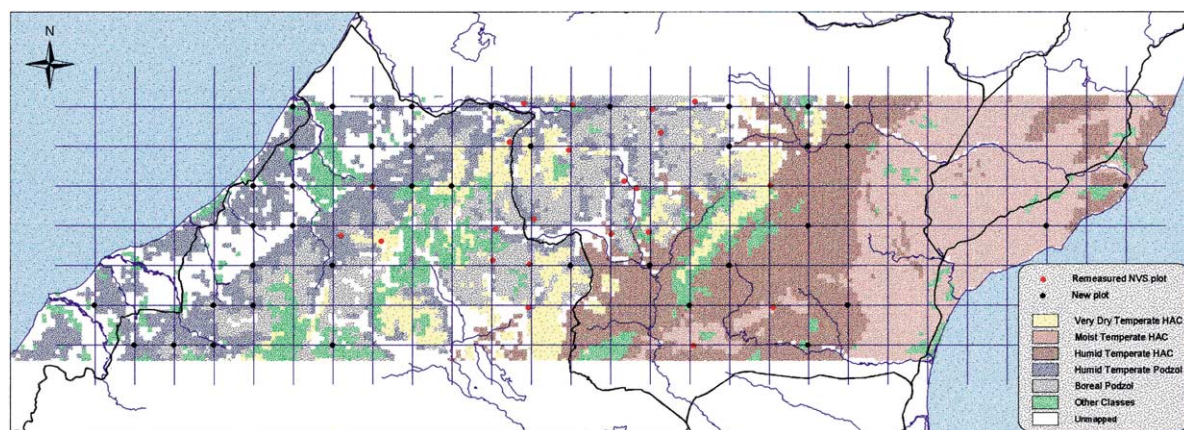


Fig. 3. Map showing major soil types along the transect, with the 9 km grid and plot locations superimposed.

Where a log protruded outside a plot, its length was taken only to the point of exit, and its diameter was measured at that point. Similarly, when a log tapered to below 10 cm diameter, its length was taken only to that point. The diameters of the trunk and major branches of standing dead trees were estimated by eye.

The volume of each log was calculated as:

$$V = \frac{\pi l}{32} ((a + b)^2 + (c + d)^2) \quad (4)$$

where the log has length l , orthogonal diameters of a and b at one end, and c and d at the other (Goulding, 1996).

The C-stock is estimated by multiplying log volume by the dead-wood density. Dead-wood density decreases considerably as logs decay, because the shape is largely retained while the log becomes increasingly hollowed by saprophytic organisms. We collated data from previous studies to estimate dead-wood densities, in the three decay classes, as a proportion of fresh-wood (FW) density: these proportions were defined as decay-stage modifiers. For each log in a plot that could be identified by species, a species-specific basic wood density (samples held at Landcare Research) was multiplied by the appropriate decay-stage modifiers to obtain a dead-wood density. Many logs were not identified by species, in which case an average FW density was used. The biomass of all CWD in the plot was calculated as:

$$\begin{aligned} \text{CWD biomass} = & \sum (\text{log volume}) \\ & \times (\text{fresh-wood density}) \\ & \times (\text{decay-stage modifier}) \end{aligned} \quad (5)$$

Previous studies show that the carbon content of rotting wood remains close to 0.5 (Allen et al., 1997), so the biomass was converted into a C-stock by multiplying by 0.5. Finally, we slope-corrected and expressed the results in t/ha.

2.6. C-stock in mineral soils and fine litter

A national soil-C monitoring system is being developed, based on default IPCC methodology (IPCC, 1996) and focusing on land-use change. This system will monitor all land-cover types, not just indigenous forests and shrublands. Soil-C stocks and fine litter were measured in some of the forest and

shrub sample plots as part of the development of a national system, and we used these data, along with other data collected in indigenous forests in the transect area, to quantify forest floor (fine material <10 cm diameter) and mineral soil-C stocks for the transect area.

Across the transect, the landscape was stratified into polygons (of variable size) defined by overlaying maps of IPCC soil categories, climate regime, and land-cover. All recognized New Zealand soil subgroups (Hewitt, 1998) were grouped into categories recommended by IPCC (1996). Climate polygons were categorized by combining two temperature and five rainfall categories, seven combinations exist in New Zealand (Leathwick and Giltrap, 1997). Twelve combinations of soil type and climate occurred in the transect area, of which soil samples were collected in the four largest areas (66% of forest and 71% of shrubland). The soil/climate polygons were then overlain on the LCDB polygons to give the area of each soil/climate/land-cover (forest or shrubland) combination in the transect area (Fig. 3).

To quantify soil-C for each soil/climate combination, we measured soil-C (bulk density and C concentration to 1 m depth) at a subset of the grid-based shrub sites and at several indigenous forest NVS plot sites in the transect area (some of which subsequently became part of the grid-based plot network). At each site, a point midway along the side of the vegetation plot was located and served as the starting point for a 20 m transect running across the slope. At 5 m intervals, we collected fine litter (<0.1 m diameter) inside 0.25 m² quadrats. Then, under each quadrat, we sampled mineral soil using stainless steel rings (0.098 m diameter by 0.1 m deep) to a depth of 0.30 m. All five samples of each 0.1 m depth increment were pooled at a site. At one point on the transect (the first one), we collected samples to 1.0 m depth or until we reached underlying rock. Soils were later air-dried, sieved (0.005 m), then analyzed for total organic C using thermal combustion (LECO FP-2000 CNS analyzer).

These data were used to calculate a mean soil (0–0.3 and 0.3–1.0 m) and fine litter C-stock for each of the major soil/climate combinations under indigenous forests and shrublands. When no data were available for a given soil/climate polygon, we used the average of all soil-C values from the transect area to estimate

the soil-C stock. We then multiplied the aerial extent of each soil/climate/land-cover (forest or shrubland) combination by the average C content to quantify the transect soil-C stock.

2.7. Additional measurements taken on plots

A number of site characteristics were quantified. These included measurements of slope, aspect, altitude, and ground-cover component (Allen, 1993), and the chemical composition of the mineral soil, determined from grab-samples taken from six points within each plot (sampled immediately beneath the humus layer to a depth of c. 10 cm). The samples were pooled, air-dried and analyzed by standard methods, for pH (0.01 M CaCl₂), available P (Bray's method), and total C and N (Leco CNS-2000 analyzer). Soil C:N ratio was calculated, as an index of soil fertility. The rainfall and potential evapotranspiration at each location were estimated from climate surfaces of New Zealand (Leathwick and Stephen, unpublished report). Water balance was defined as the yearly average of the monthly ratios of rainfall to potential evapotranspiration. The normal difference vegetation index (NDVI) of satellite imagery was derived for each of the plot locations that were accurately known. NDVI is one of the commonest vegetation indices derived from remotely sensed data, and is defined as the (NIR – RED)/(NIR + RED), where NIR and RED are the reflectance in the near infrared and red wavelengths.

2.8. Statistical methods

Generalized linear modeling was used to fit various regression relationships, for example, between C-stocks and soil available P, and to test the statistical significance of terms in these relationships (S-plus, version 4.0, Mathsoft Inc.). GLM is more powerful than least-squares regression, because non-Gaussian error structures can be applied when the variance of data is correlated to the mean (Aitken et al., 1989).

When calculating the uncertainty associated with total C-stocks, the errors associated with components were propagated using:

$$\text{var}(x + y) = \text{var}(x) + \text{var}(y) + 2 \text{covar}(x, y) \quad (6)$$

$$\frac{\text{var}(xy)}{(xy)^2} = \frac{\text{var}(x)}{x^2} + \frac{\text{var}(y)}{y^2} + \frac{2 \text{covar}(x, y)}{xy} \quad (7)$$

where $\text{var}(x)$ is the variance of x , and $\text{covar}(x, y)$ the covariance between x and y . Covariance was assumed to be zero in all calculations, and the uncertainty in aerial extents was also assumed to be zero, for lack of information otherwise.

3. Results

3.1. Ground-truthing of the land-cover database

The 43 sampling points visited for the first time provided a limited opportunity to test the spatial accuracy of the LCDB classification. There was a high misclassification rate in shrublands: of the 23 shrubland points identified by the LCDB, six were found to be forest and two were non-woody vegetation (Table 1). While the misclassification of shrubland is of concern, the random and systematic errors cannot be estimated as yet, because our sample size was small and did not include sampling points outside the area classified as forest and shrubland.

3.2. Living biomass in forest plots

The stem diameters of 6100 woody plants were measured on the transect. Over 50% of stems were sampled from just four tree species: *Nothofagus solandri* var. *cliffortioides* (mountain beech) (24%), *Weinmannia racemosa* (13%), *Quintinia acutifolia* (9%) and *Phyllocladus alpinus* (4%). An additional 79 species were measured in the plots, including trees,

Table 1
Contingency table of the classification of 62 sampling points as identified by a remotely sensed land-cover map (LCDB) and by ground-based identification

Ground-based plots	LCDB classification		Total
	Shrubland	Forest	
New plots			
Shrubland	15	1	16
Forest	6	19	25
Non-woody	2		2
Established NVS plots		18	18
Total	23	38	61 ^a

^a One forest plot was considered too dangerous to reach.

Table 2

An analysis of variance of the relationship between log (height) and log (diameter) of trees^a

Term	d.f.	MS	F-ratio	P	Model r^2
A single slope and intercept	1	228	3519	<0.001	0.55
Plus slope for each plot (common intercept)	46	106	35	<0.001	0.81
Plus intercept for each plot	46	10	3.2	<0.001	0.84
Plus slope for each species ^b	6	4.3	64	<0.001	0.85
Plus intercept for each species ^b	6	1.2	2.9	<0.001	0.86
Plus slope for each species ^b in each plot	55	7.7	2.0	<0.001	0.87
Plus intercept for each species ^b in each plot	55	1.1	0.35		0.87
Residual	976	55.1			

^a The relationships were fitted using GLM, assuming normal distributions of the residuals.^b The six most common species were included, while all other species were grouped.

forest shrubs and tree ferns, but most were very uncommon (for example, 35 of the species were recorded less than 10 times).

We tested the statistical significance of allowing the height–diameter allometric regression relationships to vary among species and among sites. The goodness of fit was greatly improved by fitting regression lines for each plot (rather than having a single line for the whole transect), and was further improved by allowing the regression lines to vary among the five most common species, with all other species pooled (Table 2).

Live tree C-stock in the 43 forest plots had a mean ($\pm$ S.E.M.) of 174 ± 29 t/ha. This estimate is slightly less than the nationwide estimate derived from 500 NVS plots in the 1990-baseline study (179 ± 5 t/ha), which included plots from a broader range of forest types (Hall et al., 1998, unpublished data). The C-stock of large trees (>600 mm dbh) in the 20 m H 20 m plot was 48.2 ± 13.8 t/ha, while that in the nested 20 m circular plot was 40.25 ± 12.1 t/ha; so sampling a larger area caused modest increase in precision.

3.3. Living biomass in shrub plots

Individual shrubs were measured on 12 of the 16 shrubland plots. A total of 2040 shrubs were measured on the 12 plots where individuals were measured (average of 157 per plot) comprising 93 species. Two early-successional shrubs, *Leptospermum scoparium* and *Coprosma propinqua*, comprised 56% of total shrub volume (Table 3). Destructive sampling produced shrub density data for only seven species, but

these species comprised 73% of the total shrub volume. These shrub species differed enormously in cuboid density (Table 3). Cover/height measurements were considered the only possible approach in four plots. For example, one of these was a subalpine shrubland, consisting of a dense mat of the recumbent conifer *Podocarpus nivalis* and intermingled larger shrubs, with very high shrub density (11.4 kg/m³). The per-hectare C-stock of the 16 shrub plots had a mean living biomass ($\pm$ S.E.M.) of 26.7 ± 7.4 t/ha. This includes the contribution made by trees (5.9 t/ha), which were found in four of the 16 shrub plots.

3.4. Coarse woody debris

Standing dead trees and logs were identified to species in 58% of cases. Most of these logs were mountain beech (36%), but another 28 species were identified. Few logs were in the early stages of decay,

Table 3

The total volumes of common shrubs across the transect, as a percentage of the total volume of all shrubs, and their cuboid densities, determined by destructive sampling

Shrub species	Total volume (%)	Density (kg/m ³)
<i>Leptospermum scoparium</i>	35	1.54
<i>Coprosma propinqua</i>	21	0.88
<i>Cytisus scoparius</i>	5	0.34
<i>Discaria toumatou</i>	5	0.86
<i>Olearia colensoi</i>	4	0.55
<i>Dracophyllum uniflorum</i>	3	3.19
<i>Carmichaelia arborea</i>	<1	1.75
Other species	27	1.17

Table 4

Decay-stage modifiers, defined as dead-wood densities as a proportion of FW density, for four key species along the transect (obtained from various studies)^a

Species	FW density	Decay-stage modifier			log (%)
		I	II	III	
<i>Nothofagus solandri</i> var. <i>cliffortioides</i>	514	0.79	0.58	0.45	36
<i>Nothofagus fusca</i>	570	0.73	0.61	0.33	1
<i>Weinmannia racemosa</i>	654	0.92	0.82	0.47	1
<i>Metrosideros umbellata</i>	800	0.75	0.64	0.62	4
All other log	490	0.82	0.66	0.47	42

^a Decomposition stages are: I, bark largely intact; II, bark and twig lost, but shape of trunk remaining intact; III, shape no longer maintained, and trunk sinking into ground.

and most were highly rotted (10, 40, and 50% of volume in the three decay classes), and of much lower density than FW (Table 4). The mean C-stock ($\pm$ S.E.M.) of CWD in forests was 28.9 ± 8.5 t/ha, while in shrubland the value was only 0.3 ± 0.3 t/ha. This reflects that much of the dead-wood in shrub plots was less than 10 cm in diameter, and that most lowland shrublands are early successional and contain little CWD.

3.5. C-stock in soil

Three soil–climate ‘types’ occupied over 65% of the area of forests and shrublands in the transect area (Fig. 3 and Table 5). In general, mineral–soil C-stocks (0–30 cm) were higher under shrublands than under forests. However, when fine litter is included, differences in total soil-C between forests and shrublands

are much smaller. A greater proportion of soil-C was found in the surface ‘O’ horizon of forests than shrublands (11 versus 6%). Overall, mineral soil and fine litter contained about 32 and 71% of total C-stocks for forests and shrublands, respectively, in the transect area (Table 6).

3.6. Scaling up from plots to the transect

Total C-stock in forests and shrublands within the transect area are given in Table 6. The C-stocks given in preceding sections were based on field-classification of what constituted forest and shrubland. However, when scaling up we have to use the map-based classification, even though it contains inaccuracies. In particular, five of the shrub sampling points were forest when visited, so LCDB-categorized shrublands have much higher carbon stocks and standard errors

Table 5

Aerial extent and per-hectare C-stocks (t/ha) of litter (diameter <10 cm) and mineral soils in five major soil classes on the transect^a

	Indigenous forests			Shrublands		
	Area (kha)	Litter	Mineral soil (0–30 cm)	Area (kha)	Litter	Mineral soil (0–30 cm)
Humid temperate podzol	120	14.8 ± 5.7 (14)	64.7 ± 6.7 (17)	34	6.6 ± 5.7 (6)	129.5 ± 26.0 (6)
Boreal podzol	45	14.5 ± 3.7 (7)	90.8 ± 12.4 (9)	22	7.5 ± 3.6^b	103.0 ± 11.4^b
Humid temperate HCA	30	20.1 ± 4.6 (4)	96.1 ± 8.0 (4)	51	15.3 ± 9.9 (8)	80.5 ± 13.5 (8)
Moist temperate HCA	13	15.5 ± 3.3^b	79.9 ± 26.2 (2)	18	1.0 ± 0.8 (7)	89.0 ± 10.6 (6)
Other soils	87	15.5 ± 3.3^b	76.6 ± 5.2^b	50	7.5 ± 3.6^b	103.0 ± 11.4^b
Total ^c	295	15.5 ± 2.0	76.1 ± 2.7	175	8.9 ± 2.2	100.2 ± 5.1

^a HCA, high clay activity.

^b The mean of all samples in the relevant category.

^c Area-weighted means and standard errors.

Table 6

Stocks ($\pm$ S.E.M.) of all major carbon components in indigenous forests and shrublands along the South Island transect, with sample sizes taken in brackets^a

Vegetation type	Per-hectare C-stocks (t/ha)					Area	Total C (million tonnes)
	Living biomass	Coarse woody debris	Fine litter ^b	Mineral soil ^b	All pools		
Indigenous forest	169.1 $\pm$ 18.4 (38)	29.6 $\pm$ 9.5 (38)	15.5 $\pm$ 2.0 (25)	76.1 $\pm$ 2.6 (32)	290.3 $\pm$ 21.0 (38)	295	85.6 $\pm$ 6.2
Shrubland	48.6 $\pm$ 13.5 (23)	5.2 $\pm$ 3.2 (23)	8.9 $\pm$ 1.1 (21)	100.2 $\pm$ 5.2 (20)	162.9 $\pm$ 14.8 (23)	175	28.5 $\pm$ 2.6
Forest + shrubland	123.6 $\pm$ 14.8 (61)	20.5 $\pm$ 7.8 (61)	12.5 $\pm$ 1.7 (46)	85.3 $\pm$ 3.8 (52)	241.9 $\pm$ 17.2 (61)	470	114.5 $\pm$ 8.1

^a These values were derived from the LCDB classification of vegetation types, and differ from those given in the text which were derived from field classifications.

^b Soils were sampled near vegetation plots and elsewhere, and have different sample sizes.

than ground-based values (biomass + CWD, 48.6 ± 13.5 versus 26.7 ± 7.4 t/ha).

The precision of total C-stock estimates can be improved by increasing the sample size of the component with largest variance—either by increasing the number of plots, or the size of plots. In both forests and shrublands, the living biomass had, by far, the largest variance, so sampling should focus on reducing the variance of this component (Table 6).

Trees <100 mm diameter comprised 67% of all stems, but only 6% of total biomass, while trees >600 mm diameter comprised 2% of all stems measured, but 36% of total biomass; clearly there is a need to reduce the variance of large-tree biomass (Fig. 4). The need to focus on large trees was incorporated to an extent in our nested plot design, with a larger plot for large trees superimposed on a smaller plot for smaller individuals. However, our data indicated that increasing the

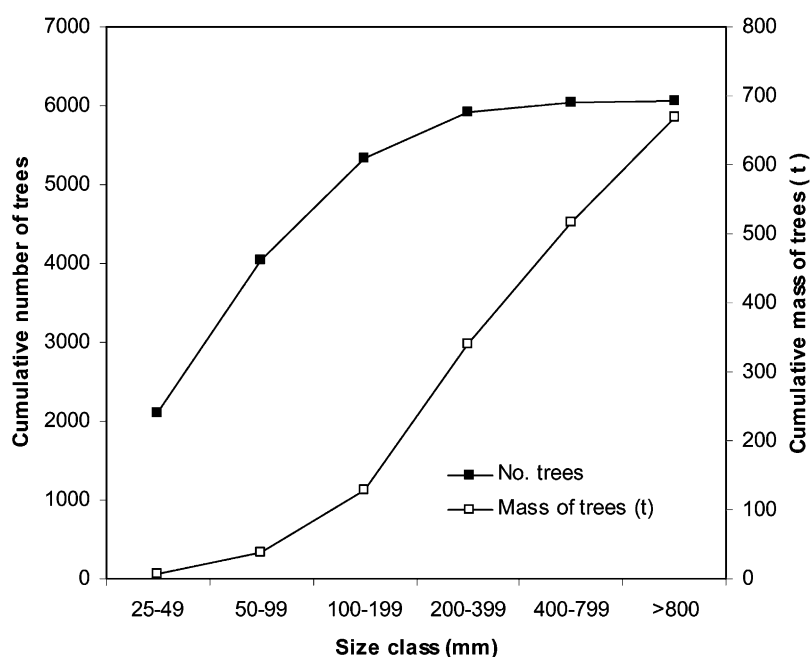


Fig. 4. Cumulative number of stems, and cumulative mass of trees (tonnes) by size class, taken from 6100 measurements on the South Island transect.

plot size did not substantially increase the precision of the estimates, perhaps because much of the variability lies between sites (e.g. different forest types). It seems likely that greater precision would come from more plots rather than larger plots.

It should be noted that uncertainty values were calculated for total C-stocks on the assumption of zero covariance among component C-stocks (Table 6). We had to make this assumption because the soil and vegetation samples were not all taken on the same plot, so we are unable to calculate covariance between soil + litter components and CWD + living biomass components. Future work needs to incorporate measurements that will allow an estimation of covariance, although these may be small. For example, we were surprised to find that C-stocks in CWD were not correlated to those in living biomass (Pearson's $r = 0.26$, $P = 0.09$).

3.7. C-sequestration in forest plots

The NVS forest plots revisited as part of the system test were first measured between 13 and 24 years previously, and provided a means of analyzing change in the living biomass pool (but not soil or CWD). Carbon sequestration in living biomass was calculated in these plots using the change-in-C-stock approach. The mean sequestration ($\pm$ S.E.M.) in living biomass

was $1.3 (\pm 0.67)$ t/ha/year (not statistically different from zero change). This change was brought about by carbon gains through tree growth of $4.4 (\pm 0.4)$ t/ha/year, minus carbon losses through mortality of $3.1 (\pm 1.2)$ t/ha/year. Power analysis indicates that 600 forest plots would detect statistically significant trends in living biomass of around 0.05 t/ha/year, equivalent to around $\pm 0.03\%$ of living biomass per year.

3.8. Predicting carbon stocks from satellite imagery and site variables

Some forest properties change systematically along environmental gradients, for example, the slopes of the height–diameter functions were significantly steeper in plots with low soil C:N ratios, $F_{1,41} = 6.4$, $P = 0.03$. Therefore we sought relationships between C-stocks (in living biomass and CWD) and site factors such as altitude, slope, water balance, soil C:N ratio, available soil P, soil pH, and a satellite-based vegetation index (NDVI). GLM was used to test for such relationships, assuming a Poisson error distribution (log-link function) to accommodate the increase in variance with mean. Very few relationships were significant. Shrubland C-stocks were correlated with NDVI calculated from satellite imagery ($r = 0.72$, $P = 0.01$) (Fig. 5), but with not other variable. Forest C-stocks were unrelated to any variables, although

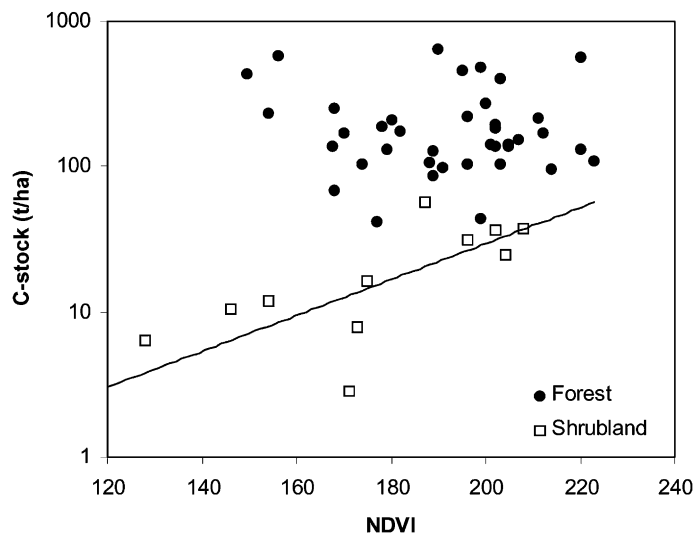


Fig. 5. Relationship between C-stocks (biomass and CWD) and a satellite-derived index of vegetation (the NDVI).

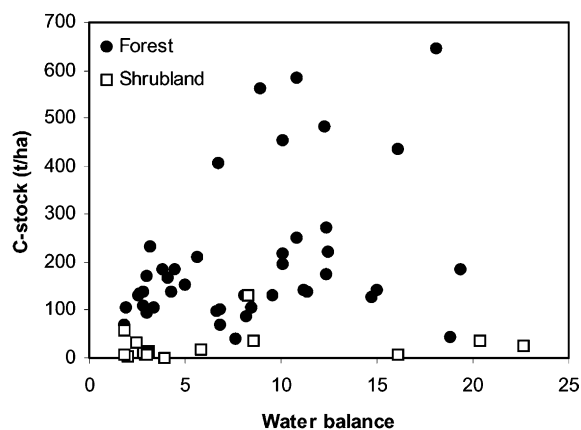


Fig. 6. Relationship between C-stocks (biomass and CWD) and estimated water balance of the transect plots.

there may have been a weak triangular relationship (Fig. 6). The fact that systematic changes in some forest properties, such as growth and tree form, do not translate into significant trends in C-stocks reflects uncertainly created by differential disturbance among plots.

4. Discussion

4.1. Design of permanent plot networks

The monitoring system described in this paper builds upon methods used to estimate C-stocks in 1990 (Hall et al., 2001), and is capable of providing unbiased and precise estimates of C-stocks. While forest inventories are routinely made around the world (Kauppi et al., 1992), our study is among the first in a refereed journal to provide regionally unbiased estimates of shrubland C-stock, and to include estimates of coarse woody debris, fine litter and soils. Our plot design has been kept simple deliberately, because we endorse the perspective of Overton and Stehman (1996) that more complex designs come with an inherent set of disadvantages, both in terms of statistical analyses, and presentation to policy makers with little interest in the fineries of sampling methodologies.

The vegetation and soil monitoring systems have different emphases. The soils system builds on IPCC

recommendations, in which pre-stratification takes place using soil, climate and vegetation information, and then C-stocks are obtained for each class. Under this system climate and soil polygons are not considered to vary over time, and carbon sequestration/emission is assumed to be driven by land-cover/land-use changes. In contrast, vegetation monitoring in permanent plots provides very precise data on temporal changes in per-hectare C-stocks, by factoring out spatial variation, while remote-sensing provides data on land-cover change. The two methodologies are convergent, at a given point in time, if the sites selected under the first approach are truly representative. Experience would indicate that the selection of 'typical' areas, by any method other than random sampling, may result in bias (Brown and Lugo, 1992). For example, previous studies have established that CWD is 24–27% of living biomass in selected New Zealand forests (Stewart and Allen, 1998), but representative sampling across the transect gives 15%, a value much closer to that obtained in the northern Hemisphere forests (Harmon and Hua, 1991).

The long-term purpose of the national system will be to monitor C-sequestration rates (i.e. changes in C-stocks per unit time), not to provide one-off data on C-stock estimates as we have done here. Not until the plots are remeasured will we know how many plots are required to estimate C-sequestration within specified confidence limits, or whether stratification by vegetation type has benefits on statistical grounds. For example, long-term monitoring indicates that some vegetation types are probably more prone to disturbance than others (e.g. Bellingham et al., 2000), so pre-stratifying into disturbance classes could, in theory, improve the precision of C-sequestration estimates. However, this stratification would be extremely difficult in practice, given the multiple agents of disturbance in New Zealand and their highly unpredictable nature (for example, cataclysmic earthquakes and volcanic eruptions). Another possibility that is being considered would be to establish plots outside the areas demarcated as forest and shrubland, especially in areas likely to revert to shrubland or forest, such as marginal agricultural land. These additional samples would allow assessment of the changes in vegetation cover that could be used alongside remote-sensing.

4.2. Multifunctionality

For brevity's sake we have not described in this paper the measurements taken on plots for purposes other than carbon reporting, but we consider these measurements as very important. Measurements of vascular-plant species composition and regeneration success were taken in all plots, using standard NVS methods (Allen, 1993). These included counts of regenerating trees and shrubs (by species) in four height classes, and a full list of vascular species, with the cover of each species estimated in six tiers. In the report upon which this paper is based (Coomes and Beets, 1999), we analyze these data to assess patterns of biodiversity, weed invasion, and dominance in different strata. For example, naturalized exotics already surpass natives in terms of numbers of species in New Zealand, and pose a huge risk to some natural ecosystems (Atkinson and Cameron, 1993). We found that most plots on the transect had no weeds whatsoever (55%), but that shrublands tended to have more exotic species than forests ($\chi^2_1 = 9.7$, $P < 0.005$), and that the number of exotic species was negatively related to water balance in both forest and shrubland ($\chi^2_1 = 24.5$ and 25.0 , $P < 0.001$). Repeat measurements on permanent forest plots are providing unique perspectives on the invasion of exotics. For example, Wiser et al. (1998) found that *Hieracium lepidulum* preferentially invades sites with higher species richness in New Zealand's mountain beech forests (250 permanent plots over 25 years), directly countering the pervasive theory that high biodiversity makes communities more resilient to invasion.

The value of historical data is often neglected in the design of new networks (Bakker et al., 1996), reducing our ability to detect deleterious human impacts (Willig et al., 1998). Perhaps, the very concept of maintaining long-term monitoring systems is anachronistic in a world where 'change' is thought synonymous with 'progress' (Bellingham et al., 2000). Quality long-term datasets tend to be put to many uses, some completely unanticipated at the time of establishment (e.g. Harcombe et al., 1998; Bellingham et al., 2000). For this reason, the proposed system incorporates established NVS plots wherever this is possible without introducing substantial bias (Hall et al., 1998, unpublished data), and include non-carbon

measurements on the plots. Clearly, these measurements take extra time (around 20%), time that could have been used to collect more carbon-related measurements and possibly improve accuracy, but we reason that the benefits far outweigh the costs. Redesign of plot networks must be evaluated within the context of the benefits of maintaining permanent plots, and their multifunctional nature.

4.3. Regression relationships

A major shortcoming of biomass regression relationships is that they are often site- and species-specific. Like many countries, New Zealand has only a few biomass relationships for native trees obtained from a handful of species and sites (Beets, 1980; Allen et al., 1998). The extrapolation of such relationships to a wider set of sites and species is often questioned (Schmitt and Grigal, 1981), but what are the alternatives, given the costs and difficulties of collecting biomass data? Are generalized formulae an achievable goal?

In order to examine this question, we have surveyed recent literature. Most biomass studies focus on single locations, and use least-squares regression to fit one of these functions to the data:

$$AGB = \alpha + \beta d$$

$$\ln(AGB) = \alpha + \beta \ln(d)$$

When biomass relationships of this sort are compiled, for example, the 143 biomass-diameter relationships from 65 North American tree species listed by Ter-Mikaelian and Korzukhin (1997), it becomes immediately apparent that the regression coefficients vary considerably among species, and vary intraspecifically among sites. The three main sources of the variation in the biomass of trees among sites and species are: (a) differences in maximum height; (b) differences in wood density; and (c) differences in form and allocation patterns. These are considered in turn in the next three paragraphs.

Obviously, some tree species grow taller than others, but there is also considerable intraspecific variation in maximum height related to soil fertility, moisture balance, exposure to wind, and stem density (Baker, 1950; Hägglund, 1981). Therefore, studies that construct general regression relationships which

include several sites have shown that height is an important explanatory variable: these include a study of *Betula papyrifera* from nine locations in North America (Schmitt and Grigal, 1981), and tropical mixed-species forests/woodlands from nine locations (Brown et al., 1989). In contrast, some single-site studies report that height is not needed as an explanatory variable, because tree heights and diameters are so closely related at a particular site that the inclusion of both terms is unnecessary. Therefore, a distinction needs to be made between studies that focus on biomass relationships within a single site, and those that make comparisons between sites (e.g. Overman, 1994; Nelson et al., 1999).

Another source of variation in tree biomass is basic wood density. For example, the wood densities of the New Zealand's native trees stored in the Landcare database varied from 380 kg/m³ for *Laurelia novae-zelandiae* to 800 kg/m³ for *Metrosideros umbellata*, with most trees falling between 400 and 600 kg/m³ (Allen et al., 1997). There can also be considerable intraspecific variation among sites. For example, the whole-tree basic density of *Pinus radiata* varies by around 25% in New Zealand, apparently in relation to mean annual temperature (Cowan, 1974). Therefore, the inclusion of wood densities into biomass relationships makes them more generalized (e.g. Uhl et al., 1988), even if destructive sampling at any one site shows their inclusion to be statistically unnecessary (e.g. Overman, 1994). We are considering coring trees on a subset of plots to obtain unbiased wood densities among species and sites, recognizing that wood archived may not be representative, especially as most large trees are hollow and therefore have lower overall density than predicted by a wood sample.

Finally, trees differ in crown structure and biomass allocation among sites and species (e.g. Usoltsev and Hoffmann, 1997), but these are very difficult to quantify in the field. The pragmatic solution is to group data into broad functional types. For example, Brown et al. (1989) grouped data from the tropics into wet, moist and dry categories, which is logical given the strong changes in tree form along tropical rainfall gradients (Walter, 1973). Schroeder et al. (1997) derived equations for conifers and broadleaves, which is also reasonable given their strongly different crown forms.

Even less is known about root allocation patterns, except that they are highly variable. The few studies available around the world give root to total biomass ratio ranging from 9 to 33% (Coomes and Grubb, 2000; also see Cairns et al., 1997). As Körner (1998) astutely notes 'dry matter allocation is extremely poorly documented for trees . . . this reflects the human fascination with electronic machinery rather than spades'. Given that allocation to roots is likely to vary greatly among sites (e.g. Beets et al., 2000), more work is needed before a figure of 20% is accepted internationally (cf. Cairns et al., 1997). Realistically, below-ground allocation will remain poorly understood for the foreseeable future.

In conclusion, studies need to build upon works such as those of Brown et al. (1989) and Schmitt and Grigal (1981), by collating data from several sites to derive general relationships for tree biomass. Height and wood density measurements are needed for generalized relationships, irrespective of whether these variables are necessary for developing within-sites regression relationships, and samples need to be taken from the range of environmental conditions to capture unbiased information on tree form.

4.4. Remote-sensing

Remote-sensing has become the only practicable method for monitoring temporal changes in land-cover over large spatial areas, and provides accurate results when the same satellite is used for repeat measurements, and imagery is interpreted using the same mathematical functions (e.g. FAO, 1996). Problems arise when land-cover maps are updated using different imagery and/or interpreted using different methods, because it is then impossible to differentiate true land-cover change from methodological artefacts. The shift from the VCM to the LCDB in New Zealand provides a case in point. The LCDB (satellite imagery) has 50 times greater spatial resolution than the VCM (photogrammetric imagery and extensive ground-based sampling), but has lower thematic resolution (2 versus 31 classes in woody indigenous vegetation). As a result of these changes, the two databases give very different land-cover estimates, for example, the VCM estimate of shrubland area is over twice that of the LCDB on the transect (43.6 versus 17.5 km²), and about 20%

became unmapped or unclassified in the soils work (Fig. 3). These differences have created obvious difficulties when trying to relate historical C-storage estimates with the LCDB-based vegetation monitoring system (Tate et al., 1997). There is a need for consistency in spatial and thematic resolution of land-cover maps for long-term, national-scale monitoring of C-stocks.

Like any sampling approach, remotely sensing has random errors associated with land-cover estimates, which can be assessed by a number of ground-truthing approaches (e.g. Cohen, 1960; Congalton, 1991; Stedman and Czaplewski, 1998). Errors arise from classifying vegetation into types, even though vegetation types such as alpine shrubland and montane forest often intergrade, and from assuming that pixels at boundaries are purely one type or another, rather than 'fuzzy', as well as relating to the skills and judgments of the analyst (Foody, 1995). For the determination of carbon stocks, the uncertainty of land-cover estimates is likely to be small compared to the plot-based uncertainty. However, temporal changes in C-stocks will be measured with high precision in permanent plots (because of repeat measurements) making random errors in land-cover estimates a real concern. More research on integrating ground-based and remotely sensed systems is needed.

In the future satellite imagery may provide reliable estimates of per-hectare C-stocks, as well as vegetation cover. Plant canopies differentially absorb red and infrared radiation (Salisbury and Ross, 1978), so NDVI values obtained from satellite imagery may correlate to vegetation cover and leaf area, especially when highly contrasting vegetation types are compared, such open woodlands with dense woodlands (e.g. De Gier and Sakouhi, 1996), and managed plantations of various ages (Tomppo, 1996b). Even then care is needed to correct for topography and solar angle, and to calibrate for water balance (Trotter, 1998; Gobron et al., 1997). However, there is generally a poor correlation between NDVI and vegetation biomass, especially when vegetation has a closed canopy and a mixture of species (e.g. our results, Trotter et al., 1997). Other remote-sensing methods, such as radar imagery (e.g. Treuhaft et al., 1996; Freeman and Durden, 1998; Pairman et al., 1999), lidar imagery (Lefsky et al., 1999) and pulse-laser profiling (S. Brown, personal communication)

may provide clearer discrimination of closed forests with differing structures, but these methods are still in their infancy. Growth modeling using multi-band spectral imagery also shows promise. For the foreseeable future though, plot-based approaches will continue to provide the only accurate method for monitoring per-hectare C-stocks.

Our ability to predict C-stocks from climate and edaphic factors also appears limited. There was no relationship between biomass and site variables for either shrubland or forest. Significant relationships between C-stocks and soil/climate data have been found in New Zealand's forests when larger datasets are examined (Harcombe et al., 1998; Hall et al., 2001), but even then the residual variation is very large. Presumably this poor predictive ability results from differences in the disturbance history of each sampling point, a particularly difficult attribute to quantify (e.g. Allen et al., 1999). For the moment, it seems that maps of C-stocks will consist of average values within broad vegetation classes.

5. Conclusions

Carbon monitoring systems need to be sufficiently simple to make reports comparable and verifiable among nations, without casting aside the efficiency benefits that may arise from stratification, nested designs and so forth. Our system is among the first to provide unbiased estimates of all major components of carbon in forests and shrublands at a given point in time. With further data collection over time, the system will be further refined to efficiently monitor national C-fluxes in indigenous forests and shrublands.

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References

- Aitken, M., Anderson, D., Francis, B., Hinde, J., 1989. Statistical Modelling in GLIM. Oxford University Press, Oxford, UK.
- Allen, R.B., 1993. A Permanent Plot Method for Monitoring Changes in Indigenous Forests. Landcare Research, Christchurch, New Zealand.
- Allen, R.B., Clinton, P.W., Davis, M.R., 1997. Cation storage and availability along a Nothofagus forest development sequence in New Zealand. *Can. J. For. Res.* 27, 323–330.
- Allen, R.B., Wiser, S.K., Hall, G., Moore, T., Goulding, C., Beets, P., Nordmeyer, A., 1998. A national system for monitoring change in carbon storage in indigenous forests and scrub: testing plot locations on a South Island transect. Unpublished Contract Report JNT9798/166 for the Ministry for the Environment, P.O. Box 10-362, Wellington, New Zealand.
- Allen, R.B., Bellingham, P.J., Wiser, S.K., 1999. Immediate damage by an earthquake to a temperate montane forest. *Ecology* 80, 708–714.
- Atkinson, I.A.E., Cameron, E.K., 1993. Human influence on the terrestrial biota and biotic communities of New Zealand. *Trends Ecol. E* 8, 447–451.
- Baker, F.S., 1950. Principles of Silviculture. 414 pp.
- Bakker, J.P., Olff, H., Willems, J.H., Zobel, M., 1996. Why do we need permanent plots in the study of long-term vegetation dynamics? *J. Veg. Sci.* 7, 147–156.
- Beets, P.N., 1980. Amount and distribution of dry matter in a mature beech/podocarp community. *New Zealand J. For. Sci.* 10, 395–418.
- Beets, P.N., Robertson, K.A., Ford-Robertson, J.B., Gordon, J., Maclaren, J.P., 2000. Description and validation of C-change: a model for simulating carbon content in managed *Pinus radiata* stands. *New Zealand J. For. Sci.* 29, 409–427.
- Bellingham, P., Wiser, S., Coomes, D., Dunningham, A., 2000. Review of Permanent Plots for long term Monitoring of New Zealand's Indigenous Forests. Science for Conservation 151. Department of Conservation, Wellington, New Zealand.
- Bouwman, A.F., 1990. Exchange of greenhouse gases between terrestrial ecosystems and the atmosphere. In: Bouwman, A.F. (Ed.), *Soils and the Greenhouse Effect*, Proceedings of the International Conference on Soils and the Greenhouse Effect. Wiley, Chichester, pp. 61–192.
- Bowman, D., 1998. The impact of aboriginal landscape burning on the Australian biota. *Tansley review No. 101. New Phys.* 140, 385–410.
- Brown, S., Lugo, A.E., 1992. Aboveground biomass estimates for tropical moist forests of the Brazilian Amazon. *Interciencia* 18, 8–18.
- Brown, S., Gaston, G., 1995. Use of forest inventories and geographic information systems to estimate biomass density of tropical forests—application to tropical Africa. *Environ. Monitoring Assessment* 38, 157–168.
- Brown, S., Gillespie, A.J.R., Lugo, A.E., 1989. Biomass estimation methods for tropical forests with applications to forest inventory data. *For. Sci.* 35, 881–902.
- Brown, S., Gillespie, A.K.R., Lugo, A.E., 1991. Biomass of tropical forests of south and southeast Asia. *Can. J. For. Res.* 21, 111–117.
- Brown, S., Schroeder, P., Birdsey, R., 1997. Aboveground biomass distribution of US eastern hardwood forests and the use of large trees as an indicator of forest development. *For. Ecol. Manage.* 96, 37–47.
- Cairns, M.A., Brown, S., Helmer, E.H., Baumgardner, G.A., 1997. Root biomass allocation in the worlds upland forests. *Oecologia* 111, 1–11.
- Cannell, M.G.R., Milne, R., 1995. Carbon pools and sequestration in forest ecosystems in Britain. *Forestry* 68, 361–378.
- Cannell, M.G.R., Cruickshank, M.M., Mobbs, D.C., 1996. Carbon storage and sequestration in the forests of northern Ireland. *Forestry* 69, 155–165.
- Catchpole, W.R., Wheeler, C.J., 1992. Estimating plant biomass: a review of techniques. *Aust. J. Ecol.* 17, 121–131.
- Cihlar, J., 2000. Land cover mapping of large areas from satellites: status and research priorities. *Int. J. Remote Sens.* 21, 1093–1114.
- Cohen, J., 1960. A coefficient of agreement for nominal scales. *Educ. Psychol. Measurement* 20, 37–46.
- Coomes, D.A., Beets, P., 1999. Monitoring carbon in indigenous forests and scrublands: a test of methods and logistics along a South Island transect. Unpublished Contract Report JNT9899/124 for the Ministry for the Environment, P.O. Box 10-362, Wellington, New Zealand.
- Coomes, D.A., Grubb, P.J., 2000. Impacts of root competition in forests and woodlands: a theoretical framework and review of experiments. *Ecol. Monogr.* 70, 171–207.
- Congalton, R.G., 1991. A review of assessing the accuracy of classifications of remotely sensed data. *Remote Sens. Environ.* 37, 35–46.
- Cowan, D.J., 1974. Wood density of radiata pine: its variation and manipulation. *New Zealand J. For.* 19, 84–92.
- Delaney, M., Brown, S., Lugo, A.E., Torreslezama, A., Quintero, N.B., 1997. The distribution of organic carbon in major components of forests located in five life zones of Venezuela. *J. Trop. Ecol.* 13, 697–708.

- Delaney, M., Brown, S., Lugo, A.E., Torreslezama, A., Quintero, N.B., 1998. The quantity and turnover of dead wood in permanent forest plots in six life zones of Venezuela. *Biotropica* 30, 2–11.
- De Gier, A., Sakouhi, A., 1996. Woody Biomass Mapping, Using Field Data and Spot Satellite Imagery. New Thrusts in Forest Inventory, Vol. I. IUFRO XX World Congress, 6–12 August 1995, Tampere, Finland. Proceedings of EFI, Vol. 7. European Forest Research Institute, Joensuu, pp. 205–211.
- De Vries, P.G., 1973. A general theory on line intersect sampling with application to logging residual recovery. *Meded. Landbouwhogeschool* 73 (11).
- Fan, S., Gloor, M., Mahlman, J., Pacala, S., Sarmiento, J., Takahashi, T., Tans, P., 1998. A large terrestrial carbon sink in North America implied by atmospheric and oceanic carbon dioxide data and models. *Science* 282, 442–446.
- FAO, 1996. Forest resources assessment 1990—survey of tropical forest cover and study of change processes. FAO Forestry Paper 130. Food and Agricultural Organisation of the United Nations, Rome, Italy.
- Foody, G.M., 1995. Fully Fuzzed Supervised Image Classification. *Remote Sensing in Action*. Remote Sensing Society, Nottingham, UK, pp. 1187–1194.
- Foody, G.M., Lucas, R.M., Curran, P.J., Honzak, M., 1997. Mapping tropical forest fractional cover from coarse spatial resolution remote sensing imagery. *Plant Ecol.* 131, 143–154.
- Foster, D.R., 1992. Land-use history (1730–1990) and vegetation dynamics in central New England, USA. *J. Ecol.* 80, 753–772.
- Freeman, A., Durden, S.L., 1998. A three-component scattering model for polarimetric SAR data. *IEEE Trans., Geosci. Remote Sens.* 36, 963–973.
- Gobron, N., Pinty, B., Verstraete, M.M., Govaerts, Y., 1997. A semi-discrete model for the scattering of light by vegetation. *J. Geophys. Res.* 102, 9431–9446.
- Goulding, C.J., 1996. Measurement of trees. In: Levack, H. (Ed.), *Forestry Handbook*. New Zealand Institute of Foresters.
- Hägglund, B., 1981. Evaluation of forest site productivity. *Forest Abstracts* 42, 515–527.
- Hägglund, B., 1985. En ny svensk riksskogstaxering. A new Swedish National Forest Survey, Report No. 35. Sveriges Lantbruksuniversitet, Inst för Skogstaxering, Umeå.
- Hall, G., Wiser, S.K., Allen, R.B., Moore, T., Beets, P., Goulding, C., 1998. Estimate of the carbon stored in New Zealand's indigenous forest and scrub vegetation for 1990. Unpublished Contract Report JNT9798/147 for the Ministry for the Environment, P.O. Box 10-362, Wellington, New Zealand.
- Hall, G., Wiser, S.K., Allen, R.B., Beets, P., Goulding, C., 2001. Strategies to estimate national carbon biomass from forest inventory data: the 1990 New Zealand baseline. *Global Change Biol.*, in press.
- Harcombe, P., Allen, R.B., Wardle, J.A., Platt, K.H., 1998. Spatial and temporal patterns in structure, biomass, growth, and mortality in a monospecific *Nothofagus solandri* var. *cliffortioides* forest in New Zealand. *J. Sust. For.* 6, 313–345.
- Harmon, M.W., Hua, C., 1991. Coarse woody debris dynamics in two old-growth ecosystems. *BioScience* 41, 604–610.
- Hewitt, A.E., 1998. New Zealand soil classification. Landcare Research Report Series No. 1, 2nd Edition. Landcare Research, Dunedin, New Zealand.
- IPCC, 1996. Land-use change and forestry. In: Intergovernmental Panel for Climate Change, pp. 5.6–5.75.
- Kauffmann-Zeh, A., 1999. Resources lacking to save Amazon biodiversity. *Nature* 398, A20–A22.
- Kauppi, P.E., Mielikäinen, K., Kuusela, K., 1992. Biomass and carbon budget of European forests, 1971 to 1990. *Science* 256, 70–74.
- Kolchugina, T.P., Vinson, T.S., 1995. Role of Russian forests in the global carbon balance. *Am. Biol. J.* 24, 258–264.
- Körner, C., 1998. A re-assessment of high elevation treeline positions and their explanation. *Oecologia* 115, 445–459.
- Laurance, W.F., 1998. A crisis in the making: responses of Amazonian forests to land use and climate change. *Tree* 13, 411–413.
- Leathwick, J.R., Giltrap, D.J., 1997. Contribution of soil carbon to New Zealand's CO₂ emissions. III. Climate information. Unpublished Contract Report LC9798/002 for the Ministry for the Environment, P.O. Box 10-362, Wellington, New Zealand.
- Lefsky, M.A., Cohen, W.B., Acker, S.A., Parker, G.G., Spies, T.A., Harding, D., 1999. Lidar remote sensing of the canopy structure and biophysical properties of Douglas-fir western hemlock forests. *Remote Sens. Environ.* 70, 339–361.
- Malhi, Y., Grace, J., 2000. Tropical forests and atmospheric carbon dioxide. *Trends Ecol. E* 15, 332–337.
- Nelson, B.W., Mesquita, R., Pereira, J.L.G., de Souza, S.G.A., Batista, G.T., Couto, L.B., 1999. Allometric regressions for improved estimate of secondary forest biomass in the central Amazon. *For. Ecol. Manage.* 117, 149–167.
- Newsome, P.F.J., 1987. The vegetative cover of New Zealand. Water and Soil Miscellaneous Publication No. 112. National Water and Soil Conservation Authority, Wellington.
- Overman, J.P.M., 1994. Evaluation of regression models for above-ground biomass determination in Amazon rainforest. *J. Trop. Ecol.* 10, 207–219.
- Overton, W.S., Stehman, S.V., 1996. Desirable design characteristics for long-term monitoring of ecological variables. *Environ. Ecol. Stat.* 3, 349–361.
- Pairman, D., McNeill, S.J., Scott, N., Belliss, S.E., 1999. Vegetation identification and biomass estimation using AIR-SAR data. *Geocarto Int.* 14 (3), 67–75.
- Phillips, O.L., 1996. Long-term environmental change in tropical forests: increasing tree turnover. *Environ. Cons.* 23, 235–248.
- Phillips, O.L., Malhi, Y., Higuchi, N., Laurance, W.F., Nunez, P.V., Vasquez, R.M., Laurance, S.G., Ferreira, L.V., Stern, M., Brown, S., Grace, J., 1998. Changes in the carbon balance of tropical forests—evidence from long-term plots. *Science* 282, 439–442.
- Saatchi, S.S., Rignot, E., 1997. Classification of Boreal forest cover types using SAR images. *Remote Sens. Environ.* 60, 270–281.
- Salisbury, F.B., Ross, C.W., 1978. *Plant Physiology*, Second Edition. Wadsworth, Belmont, CA, USA.
- Sarmiento, J.L., Lequere, C., Pacala, S.W., 1995. Limiting future atmospheric carbon dioxide. *Global Biogeochem. Cycles* 9, 121–137.

- Schmitt, M.D.C., Grigal, D.F., 1981. Generalized biomass estimation equations for *Betula papyrifera* marsh. Can. J. For. Res. 11, 837–840.
- Schroeder, P., Brown, S., Mo, J.M., Birdsey, R., Cieszewski, C., 1997. Biomass estimation for temperate broadleaf forests of the United States using inventory data. For. Sci. 42, 424–434.
- Skole, D.C., Tucker, C., 1993. Tropical deforestation and habitat fragmentation in the Amazon: satellite data from 1978–1988. Science 260, 1905–1910.
- Stedman, S.V., Czaplewski, R.L., 1998. Design and analysis for thematic map accuracy assessment: fundamental principles. Remote Sens. Environ. 64, 331–344.
- Steininger, M.K., 2000. Satellite estimation of tropical forest above ground biomass: data from Brazil and Bolivia. Int. J. Remote Sens. 21, 1139–1157.
- Stewart, G.H., Burrows, L.E., 1994. Coarse woody debris in old-growth temperate beech (*Nothofagus*) forests of New Zealand. Can. J. For. Res. 24, 1989–1996.
- Stewart, G.H., Allen, R.B., 1998. The importance of woody debris in New Zealand's indigenous forests. New Zealand For. Sci. 43, 31–33.
- Tate, K.R., Giltrap, D.J., Claydon, J.J., Newsome, P.F., Atkinson, I.A.E., Taylor, M.D., Lee, R., 1997. Organic carbon stocks in New Zealand's terrestrial ecosystems. J. R. Soc. New Zealand 27, 315–335.
- Ter-Mikaelian, M.T., Korzukhin, M.D., 1997. Biomass equations for 65 North American tree species. For. Ecol. Manage. 97, 1–24.
- Tomppo, E., 1996a. Biodiversity monitoring in Finnish forest inventories. Assessment of biodiversity for improved forest management. In: Proceedings of the International Workshop on EFI, Vol. 6, 12–17 June 1995, European Forest Research Institute, Joensuu.
- Tomppo, E., 1996b. Multi-source national forest inventory of Finland. New thrusts in forest inventory. In: Proceedings of the IUFRO XX World Congress, Vol. I (7), 6–12 August 1995, Tampere, Finland. European Forest Research Institute, Joensuu.
- Treuhaft, R.N., Madsen, S.N., Moghaddam, M., van Zyl, J.J., 1996. Vegetation characteristics and underlying topography from interferometric radar. Radio Sci. 31, 1449–1485.
- Trotter, C.M., 1998. Characterising the topographic effect at red wavelengths using a juvenile conifer canopy. Int. J. Remote Sens. 19, 2215–2221.
- Trotter, C.M., Dymond, J.R., Goulding, C.J., 1997. Estimation of timber volume in a coniferous plantation forest using landsat TM. Int. J. Remote Sensing 18, 2209–2223.
- Uhl, C., Buschbacher, R., Serrão, E.A.S., 1988. Abandoned pastures in eastern Amazonia. J. Ecol. 76, 663–681.
- Usoltsev, V.A., Hoffmann, C.W., 1997. Combining harvest sample data with inventory data to estimate forest biomass. Scand. J. For. Res. 12, 273–279.
- Van Wagner, C.E., 1968. The line intersect method in forest fuel sampling. For. Sci. 14, 20–26.
- Vitousek, P.M., D'Antonio, C.M., Loope, L.L., Rejmanek, M., Westbrooks, R., 1997. Introduced species—a significant component of human-caused global change. New Zealand J. Ecol. 21, 1–16.
- Walter, H., 1973. Vegetation of the Earth. English Universities Press, London, UK.
- Wardle, P., 1991. The Vegetation of New Zealand. Cambridge University Press, Cambridge, UK.
- Warren, W.G., Olsen, P.F., 1964. A line intersect method for assessing logging waste. For. Sci. 10, 267–276.
- Willig, M.R., Secrest, M.F., Cox, S.B., Cary, J.F., Alvares, J., Gannon, M.R., 1998. Long-term monitoring of snails in the Luquillo experimental forest of Puerto Rico: heterogeneity, scale, disturbance, and recovery. In: Dallmeier, F., Comiskey, J.A. (Eds.), Forest Biodiversity in North, Central and South America, and the Caribbean: Research and Monitoring, Vol. 21. Man and the Biosphere Series. UNESCO, Paris and Parthenon Publishing, Carnforth, Lancs, UK, pp. 293–322.
- Wiser, S.K., Allen, R.B., Clinton, P.W., Platt, K.H., 1998. Community structure and forest invasion by an exotic herb over 24 years. Ecology 79, 2071–2081.
- Wiser, S.K., Bellingham, P.J., Burrows, L., 2001. Managing biodiversity information: development of the National Vegetation Survey Database. New Zealand J. Ecol., in press.